SynUW-VIBench: Underwater Visual-Inertial Benchmark with Frame and Event Cameras

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***Abstract*—Underwater visual-inertial navigation is degraded by low light, scattering, texture loss, and strong dynamic range variation, but these effects stress frame and event cameras in different ways. We present SynUW-VIBench, a Stonefish-based synthetic underwater benchmark with synchronized stereo frames, co-located stereo events, IMU, and simulator ground truth. The benchmark evaluates frame-only, frame-inertial, RGB-event, and frame-event-inertial pipelines on Stone and Coral scenes under clear, scattering-dominated blur, dark, and HDR conditions. We report trajectory error, rotation error, and failure modes. The results reveal different failure patterns across visibility conditions under the tested pipeline configurations. OKVIS2 obtains low errors in clear and blur synthetic cases, while ESVIO completes all tested synthetic dark and HDR sequences with the lowest reported ATE. AquaticVision validation further shows that selected valid runs can be competitive, but real sequences still expose run-to-run instability and divergence. SynUW-VIBench provides a diagnostic benchmark for examining frame- and event-based odometry under controlled underwater visibility and studying transfer to real environments.**

***Index Terms*—underwater SLAM, event camera**

I. Introduction

Visual-inertial odometry (VIO) is essential for underwater robots operating near reefs, infrastructure, and other GPS-denied environments. Underwater visibility is shaped by multiple factors: low light weakens image signals, motion and exposure blur textures, while scattering and turbidity degrade contrast and brightness consistency. As a result, a sensor that works well in one condition may fail in another.

Event cameras are promising for underwater navigation because they report brightness changes asynchronously with high temporal resolution and high dynamic range. This has enabled robust state estimation in challenging robotics scenarios, including event-based stereo visual-inertial odometry [1], [2]. However, underwater scattering may trigger events that do not correspond to stable geometric edges, making it necessary to evaluate when event streams are truly useful.

Recent underwater datasets such as AquaticVision provide real frame and event data with motion-capture ground truth and reveal realistic SLAM failures [3]. While real data are indispensable, they rarely isolate which visibility factor causes failure. Simulation can address this gap, but only when the sensor streams, conversion pipeline, and evaluation protocol are clearly documented for reproducible comparison.

We present SynUW-VIBench, a synthetic underwater visual-inertial benchmark for comparing frame and event cameras under controlled visibility conditions. Built on Stonefish [4], the pipeline generates stereo frames, stereo events, IMU data, and ground truth across clear, scattering-dominated blur, dark, and HDR scenes. We evaluate VINS-Fusion [5], ORB-SLAM2 [6], RAMPVO [7], OKVIS2 [8], and ESVIO [1].

This paper makes four contributions:

- We provide a synthetic underwater benchmark pipeline with stereo frames, stereo events, IMU, ground truth, and controlled visibility profiles.
- We compare stereo-only, stereo-inertial, RGB-event, and stereo-event-inertial pipelines using SE(3)-aligned ATE RMSE, rotational error, failure labels, and ESVIO repeated-run statistics.
- We show that performance is condition dependent: OKVIS2 performs well in several clear and blur synthetic cases, while ESVIO achieves the lowest reported ATE in the tested dark and HDR cases.
- We contextualize the synthetic results on AquaticVision with local RAMPVO, ESVIO, and OKVIS2 evaluations, including repeat statistics and divergence labels.

II. Related Work

A. Underwater Image Formation and Benchmarks

Underwater visual navigation differs from terrestrial SLAM because image formation is strongly affected by absorption, scattering, artificial lighting, and water path length. Marine optics and underwater imaging models show that wavelength-dependent attenuation and forward/backscatter reduce contrast and distort image structure [9], [10]. For odometry, these effects are not simple noise: they may remove texture, create non-geometric brightness changes, and affect frame and event cameras differently.

Real underwater datasets capture these challenges in operational conditions. CADDY provides underwater stereo data for diver-robot interaction, and AQUALOC combines visual, inertial, and pressure measurements for localization [11], [12]. AquaticVision further provides synchronized event and frame cameras with motion-capture ground truth and reports failures under scanning, crossing, dark, HDR, and blur conditions [3]. While such datasets expose realistic calibration, synchronization, texture, and initialization issues, they rarely isolate which visibility factor causes a failure.

This work was supported in part by the Science Center Program of the National Natural Science Foundation of China under Grant 62188101.

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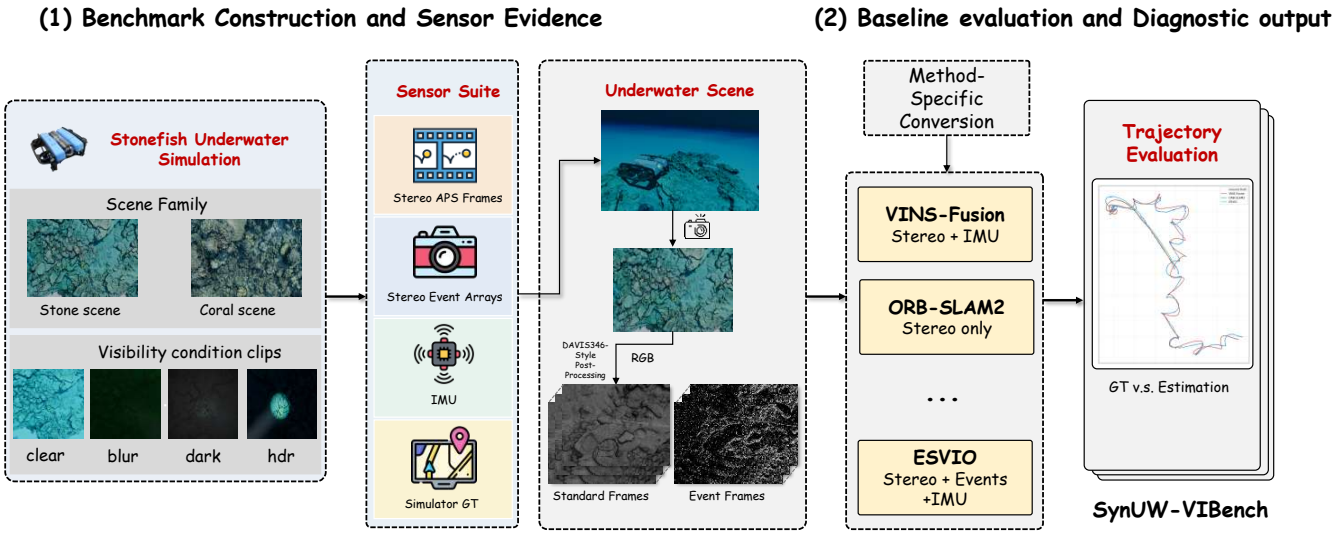


Fig. 1. SynUW-VIBench benchmark pipeline. The left side summarizes benchmark construction and sensor evidence: Stonefish scene generation, controlled visibility profiles, stereo frames, stereo event arrays, IMU, simulator ground truth, and DAVIS346-style post-processing. The right side shows method-specific conversion paths for VINS-Fusion, ORB-SLAM2, RAMPVO, OKVIS2, and ESVIO, followed by trajectory evaluation against simulator ground truth.

B. Simulation for Marine Robotics

Marine robotics simulation provides repeatable ground truth and controlled environmental settings. Stonefish is an open-source marine robotics simulator and has been extended toward machine-learning-oriented workflows [4], [13]. We use this controllability to vary scene family and visibility profile under a fixed evaluation protocol. Synthetic results are used for diagnostic analysis, while AquaticVision serves as a real-data sanity check.

III. SynUW-VIBench Benchmark Design

SynUW-VIBench is designed to expose underwater visibility failure modes under controlled sensor conditions. The synthetic environment is built in Stonefish, a marine robotics simulator, with a BlueROV2-style underwater platform. The Stone and Coral scene meshes are assembled from underwater 3D assets in the TritoniaScientific Sketchfab collection and used under the Creative Commons Attribution-NonCommercial license [14], [15]. We use two scene families, Stone and Coral, and evaluate each under clear, scattering-dominated blur, dark, and HDR profiles.

Table I compares brightness statistics of the synthetic profiles with AquaticVision. HDR shows the closest agreement in these statistics: synthetic HDR has 17.83–20.62 DN mean intensity, 41.68–47.01 DN standard deviation, and 82.3–83.1% dark pixels, compared with 17.98 DN, 43.73 DN, and 83.5% for AquaticVision S08. Dark scenes reach 100.0% dark pixels in simulation and 97.7% for S06–S07. In clear and blur scenes, the synthetic images have lower contrast and more uniform illumination. These measurements provide a quantitative reference for interpreting the four visibility profiles alongside real underwater recordings.

TABLE I
VISIBILITY STATISTICS FOR SYNTHETIC PROFILE A AND AQUATICVISION REFERENCES. MEAN/STD. ARE GRAYSCALE DN; DARK IS PIXELS BELOW 30 DN.

Scene / group	Cond.	Mean	Std.	Dark %
Stone	clear	74.68	29.72	8.2
Stone	blur	37.31	10.31	23.5
Stone	dark	2.58	3.11	100.0
Stone	hdr	20.62	47.01	82.3
Coral	clear	65.46	33.98	15.1
Coral	blur	37.68	9.86	20.1
Coral	dark	1.71	2.28	100.0
Coral	hdr	17.83	41.68	83.1
AquaticVision grouped references				
AV S01–S05	clear	77.69	49.95	19.8
AV S09	blur	50.65	29.86	26.1
AV S06–S07	dark	6.15	8.77	97.7
AV S08	hdr	17.98	43.73	83.5

A. Sensor Suite

The simulated sensor suite includes stereo active-pixel frames and co-located stereo event streams at a DAVIS346-inspired resolution of 346×260, together with IMU measurements and simulator odometry ground truth. The stereo baseline is $0.122\,\mathrm{m}$. Ground truth is recorded from the simulator odometry at $120\,\mathrm{Hz}$. Stonefish event-based cameras are updated at $1000\,\mathrm{Hz}$, corresponding to a 1 ms simulation step, and their outputs are stored as timestamped event arrays.

We describe the event stream with the standard contrast-threshold model. Let $L(\mathbf{x}, t) = \log I(\mathbf{x}, t)$ denote log intensity. An event $e_k = (\mathbf{x}_k, t_k, p_k)$ is generated when the brightness change at a pixel crosses the contrast threshold,

$$p_k\big(L(\mathbf{x}_k, t_k) - L(\mathbf{x}_k, t_k^-)\big) \geq C, \quad p_k \in \{-1, +1\}, \qquad (1)$$

where t_k^- is the previous event time at the same pixel. In our Stonefish configuration, the positive and negative thresholds

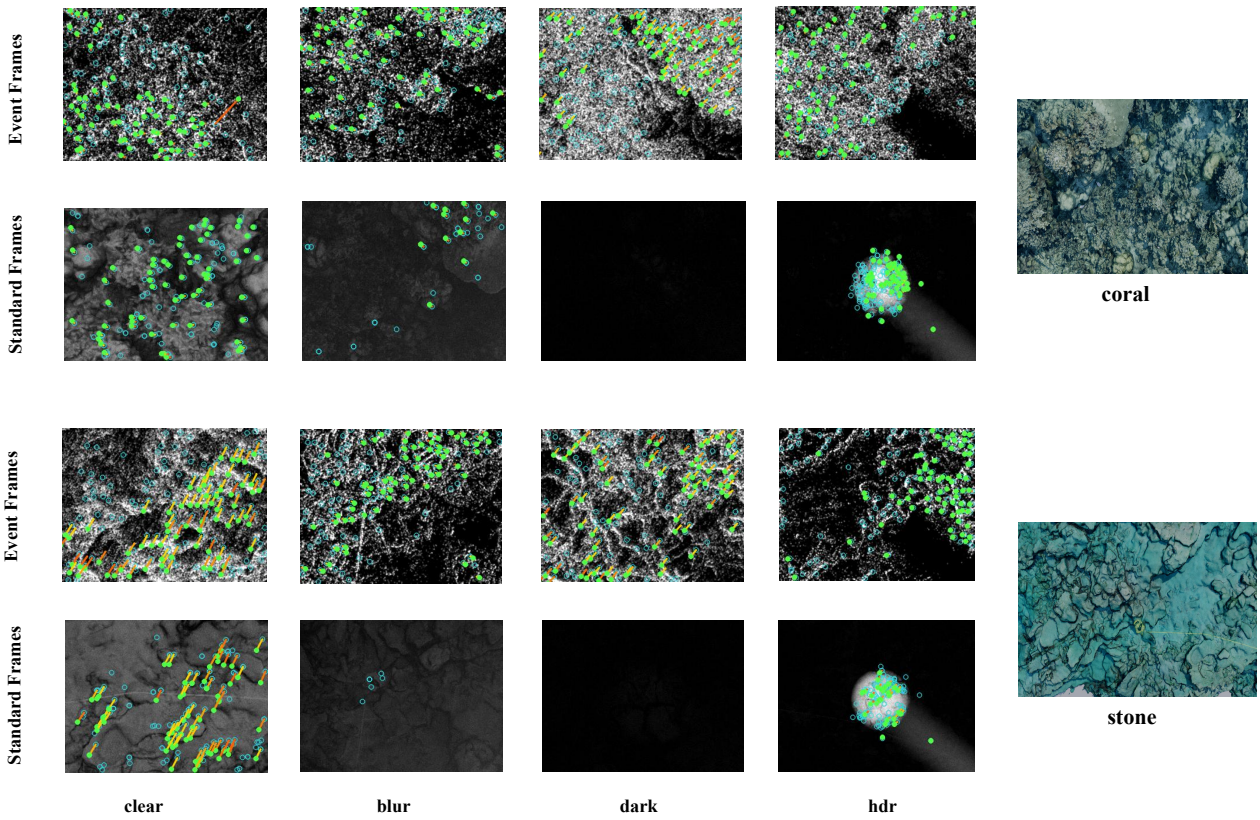


Fig. 2. Sensor-level visualization of the synthetic scenes. Standard frames and accumulated event frames are shown for clear, scattering-blur, dark, and HDR conditions in both Coral and Stone scenes. The blur cases strongly reduce standard-frame contrast and produce sparse or unstable event structure, dark cases contain weak standard-frame appearance but can still generate motion-aligned event activity, and HDR cases introduce strong local intensity variation. This qualitative view complements the brightness statistics in Table I.

are both $C = 0.28$, with a 10 ms refractory period and contrast-threshold noise of 0.01. These parameters define the event response used throughout the synthetic benchmark.

Two recording profiles are used in the reported evaluation. Profile A records 20 Hz stereo frames, 200 Hz IMU measurements, and the simulated event stream, and is used for VINS-Fusion, RAMPVO, OKVIS2, and ESVIO through method-specific sensor subsets. RAMPVO uses a monocular RGB-event subset, while ESVIO consumes stereo frames, stereo events, and IMU measurements. Profile B records 30 Hz stereo frames and 1000 Hz IMU measurements, and is used for ORB-SLAM2, although ORB-SLAM2 uses only the extracted stereo images. Both profiles are generated from the same scene, visibility condition, and simulator trajectory. Results are reported for these method-specific input configurations.

B. Visibility Profiles

The clear condition uses low turbidity and high solar elevation. The blur condition increases the Jerlov water parameter to induce scattering-dominated low contrast. The dark condition lowers the solar elevation and uses a dim downward spotlight. The HDR condition introduces stronger local intensity variation to stress dynamic-range handling rather than simply lowering global brightness. These profiles are intended to separate common underwater regimes: low-light scenes where event cameras may retain useful temporal contrast, scattering scenes where event activity can be dominated by unstable texture and backscatter rather than persistent geometry, and HDR-like scenes where local exposure variation stresses the pipelines differently.

The two scene families and four visibility profiles provide a controlled setting for examining how complete sensing pipelines respond to visibility changes, with trajectory accuracy and completion evaluated against simulator ground truth.

C. Artifacts and Reproducibility

Upon acceptance, we will release the synthetic generation environment, the recorded benchmark bags, and the conversion scripts used to prepare the data for VIO evaluation. The released recordings will cover all scene, visibility, and sensor-profile combinations described above, and the generation environment will link to the patched Stonefish branch used to produce the high-rate event-camera streams. The release metadata will preserve the Sketchfab asset attribution and license notice, so reuse of the scene assets remains subject to the original non-commercial license terms.

These planned artifacts will support inspection, reuse, and reprocessing of the synthetic benchmark data. Reproducing

the exact baseline numbers additionally requires installing the evaluated odometry pipelines and following the evaluation protocol described in the paper.

IV. Baselines and Evaluation Protocol

We evaluate five representative pipelines. VINS-Fusion is a stereo visual-inertial baseline from the VINS multi-sensor optimization framework [5]. It consumes synchronized stereo images and IMU measurements. ORB-SLAM2 is a stereo frame-only SLAM baseline [6]. It consumes synchronized stereo image pairs and does not use IMU. RAMPVO is a learning-based monocular RGB-event visual odometry baseline [7]. It consumes monocular RGB frames and event data. OKVIS2 is a stereo visual-inertial SLAM baseline with loop-closure support [8]. ESVIO is a stereo event-based visual-inertial odometry system that combines standard images, event streams, and inertial measurements [1].

For evaluation, each simulated sequence is first stored in a common timestamped sensor-stream representation. We then repackage these streams to match each baseline's runtime interface, without exposing ground-truth poses to the estimators. VINS-Fusion and OKVIS2 are run from stereo+IMU streams, ORB-SLAM2 from a timestamp-indexed stereo image sequence, RAMPVO from monocular RGB frames and event data, and ESVIO from a merged stereo+IMU+event stream using the event-array format expected by the estimator. For event-based baselines, event-array messages outside the common recording interval are removed to avoid event-only startup and shutdown segments.

For synthetic experiments, failure is reported when a method does not produce any usable trajectory for ATE evaluation. A diverged trajectory is reported separately when the method produces poses but the estimate is not meaningful for comparison. For ESVIO, we repeat each synthetic condition three times to characterize variation within that condition. When three complete valid repeated runs are available, the reported ESVIO value is the median over the three runs. If no run produces any usable trajectory, the condition is marked as failure. Partial trajectories are explicitly marked, reported only as references, and are not counted as complete successful runs. Bold values in the synthetic benchmark table mark the best valid values at the reported precision.

For AquaticVision, RAMPVO, ESVIO, and OKVIS2 are evaluated locally on the available image, event, and IMU streams required by each method. VINS-Stereo, ORB-SLAM2, and ESVO2 values are external Table III values from AquaticVision [3]. ESVIO is run three times per sequence, and the ESVIO value in the comparison table is the median ATE over complete valid runs. If only one complete valid run is available, that run is reported. Failed and diverged repeats are excluded from the scalar table value but are shown explicitly in Fig. 4. RAMPVO and OKVIS2 results are reported with the same success, failure, and divergence labels.

V. Results

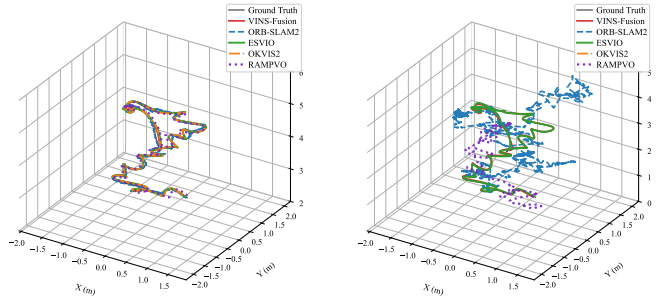
A. Synthetic Underwater Benchmark

Table II reports the main synthetic benchmark results. In the clear scenes, stereo-inertial and stereo-event-inertial methods achieve accurate trajectories, showing that these cases are not failure regimes for frame-inertial VIO. On Stone clear, ESVIO obtains the lowest ATE of 0.0070 m, followed by OKVIS2, VINS-Fusion, and RAMPVO. On Coral clear, OKVIS2 is best with 0.0036 m ATE, followed by ESVIO and VINS-Fusion. As shown in Fig. 3, the Stone clear trajectories largely overlap with the ground truth, and the differences are mainly quantitative.

ESVIO achieves the lowest reported ATE in both synthetic dark scenes. On Stone dark, VINS-Fusion diverges, ORB-SLAM2 fails, and OKVIS2 does not produce a usable trajectory, while ESVIO remains valid with 0.0651 m ATE and 5.49 degrees rotation error. RAMPVO also remains finite but with much larger error. On Coral dark, ESVIO reduces ATE from 0.1435 m to 0.0937 m compared with VINS-Fusion, and also lowers rotation error from 12.10 to 6.60 degrees.

HDR scenes separate the methods more clearly. On Stone HDR, VINS-Fusion and OKVIS2 diverge, ORB-SLAM2 and RAMPVO have high error, and ESVIO remains valid with 0.0305 m ATE and 1.73 degrees rotation error. On Coral HDR, both VINS-Fusion and ESVIO complete the trajectory, with ESVIO achieving the lowest ATE of 0.0201 m, while ORB-SLAM2 and RAMPVO drift and OKVIS2 diverges. The trajectory overlay in Fig. 3 illustrates that successful completion already becomes a key differentiator under strong intensity variation.

The synthetic results show a condition-dependent ranking under the tested configurations. OKVIS2 obtains low errors in clear and blur cases, ESVIO completes all tested dark and HDR sequences with the lowest reported ATE, RAMPVO is competitive in selected regimes, and ORB-SLAM2 has large errors or tracking failures under degraded visibility. The real-data comparison examines how these patterns carry over to AquaticVision sequences.



(a) Stone clear (b) Coral HDR

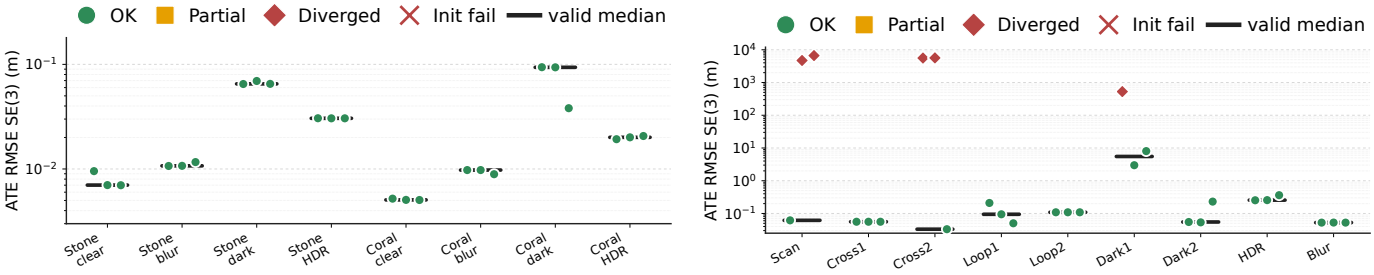
Fig. 3. Representative 3D trajectory overlays. Stone clear shows nominal tracking, where the strongest pipelines closely follow the ground truth. Coral HDR is more challenging: VINS-Fusion and ESVIO complete the trajectory, RAMPVO remains finite but inaccurate, ORB-SLAM2 drifts substantially, and OKVIS2 diverges under local evaluation.

TABLE II

SYNTHETIC BENCHMARK RESULTS. VALUES ARE SE(3)-ALIGNED ATE RMSE IN METERS AND ROTATION RMSE IN DEGREES; LOWER IS BETTER. BOLD MARKS THE BEST VALID VALUES AT THE REPORTED PRECISION. ESVIO REPORTS MEDIANS OVER THREE COMPLETE RUNS WHEN AVAILABLE.

	Stone								Coral							
	Clear		Blur		Dark		HDR		Clear		Blur		Dark		HDR	
Method	ATE	Rot.	ATE	Rot.	ATE	Rot.	ATE	Rot.	ATE	Rot.	ATE	Rot.	ATE	Rot.	ATE	Rot.
VINS-Fusion	0.0089	0.34	0.0180	0.83	*div.*	*div.*	*div.*	*div.*	0.0054	0.27	0.0147	0.80	0.1435	12.10	0.0355	2.01
ORB-SLAM2	0.0227	0.79	0.0430	1.17	*fail*	*fail*	1.5858	40.14	0.0106	0.66	0.0401	1.10	*fail*	*fail*	0.9327	75.59
RAMPVO	0.0146	0.39	0.0086	**0.29**	1.0813	150.79	0.6869	168.66	0.0169	0.49	0.0201	0.53	1.4094	135.24	0.7603	87.55
OKVIS2	0.0080	**0.19**	**0.0065**	0.37	*fail*	*fail*	*div.*	*div.*	**0.0036**	**0.21**	**0.0062**	**0.37**	10.3273	54.51	*div.*	*div.*
ESVIO	**0.0070**	0.37	0.0107	0.34	**0.0651**	**5.49**	**0.0305**	**1.73**	0.0051	**0.21**	0.0098	0.56	**0.0937**	**6.60**	**0.0201**	**1.03**

fail: no valid trajectory was produced; *div.*: diverged trajectory.



(a) Synthetic ESVIO repeats

(b) AquaticVision ESVIO repeats

Fig. 4. ESVIO repeated-run stability. Each point is one run, marker shape indicates run status, and the horizontal segment marks the median over valid runs. Synthetic repeats are valid across clear, blur, dark, and HDR conditions, with larger errors in dark and HDR than in clear and blur. AquaticVision repeats show that selected valid runs can be competitive, but Cross1, Loop2, Dark1, and Blur still expose divergence. Divergence markers indicate failed estimates, not low-error trajectories.

TABLE III

AQUATICVISION VALIDATION RESULTS. VALUES ARE SE(3)-ALIGNED ATE RMSE IN METERS; LOWER IS BETTER. AQUATICVISION BASELINES ARE REPORTED FROM THE ORIGINAL BENCHMARK; RAMPVO, ESVIO, AND OKVIS2 ARE EVALUATED UNDER OUR LOCAL PROTOCOL, WITH ESVIO REPEAT STABILITY SHOWN IN FIG. 4.

Method	S01 Scan	S02 Cross1	S03 Cross2	S04 Loop1	S05 Loop2	S06 Dark1	S07 Dark2	S08 HDR	S09 Blur
VINS-Stereo	0.057	0.062	0.043	0.049	0.091	1.163	0.070	*fail*	0.051
ORB-SLAM2	0.423	0.113	0.086	0.427	0.171	*fail*	*fail*	0.176	0.155
RAMPVO	0.103	0.058	0.046	0.092	0.094	0.437	0.110	0.374	0.065
OKVIS2	0.063	*div.*	0.048	0.089	*div.*	*div.*	0.064	0.141	*div.*
ESVIO	0.062	0.056	0.033	0.050	0.109	2.981	0.054	0.253	0.053
ESVO2	*fail*	*fail*	*fail*	*fail*	*fail*	*fail*	*fail*	*fail*	*fail*

fail: no valid trajectory was produced; *div.*: diverged trajectory.

B. AquaticVision Real-Data Validation

Table III reports AquaticVision validation with external AquaticVision baselines and locally evaluated RAMPVO, ESVIO, and OKVIS2 entries. ESVIO can be competitive on several real underwater sequences: among the reported methods, it achieves the lowest ATE on Cross1, Cross2, and Dark2. RAMPVO is competitive on Cross1 and is notably stronger than the other locally evaluated methods on Dark1, but it is less accurate than ESVIO on several other sequences. OKVIS2 is valid on Scan, Cross2, Loop1, Dark2, and HDR, but diverges on Cross1, Loop2, Dark1, and Blur under the local evaluation. The repeat set in Fig. 4 remains essential for interpreting ESVIO: even when selected valid runs are accurate, real sequences can still show large run-to-run variation and divergence markers.

The real-data comparison complements the controlled synthetic experiments. RAMPVO, ESVIO, and OKVIS2 each achieve accurate tracking on selected real DAVIS underwater sequences, with distinct patterns of error and failure across conditions. Reporting both trajectory error and run status makes these differences visible across methods and sequences.

VI. DISCUSSION

A. Performance Across Visibility Conditions

ESVIO's low errors and completed trajectories in the synthetic dark and HDR cases are consistent with the ability of event cameras to retain motion cues under challenging illumination [16]. Frame-inertial pipelines, especially OKVIS2, also obtain low errors in clear and blur scenes. Together, these observations show that the tested visibility regimes challenge the evaluated pipelines in different ways.

The benchmark compares complete sensing pipelines with their native inputs and estimation strategies. Each method's

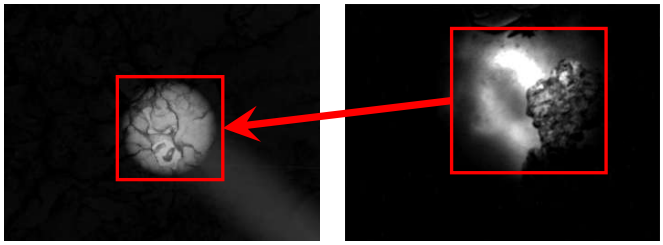


Fig. 5. Synthetic and real underwater illumination. The simulated scene has smoother localized lighting, while the AquaticVision frame combines spotlight, shadow, backscatter, and object reflectance.

behavior reflects the joint effect of its sensor configuration and estimator design under the tested visibility conditions.

B. Scattering and Learned Representations

In the synthetic blur scenes, OKVIS2 gives the lowest ATE, RAMPVO gives the lowest rotation error on Stone blur, ESVIO remains competitive, and ORB-SLAM2 is less accurate than the best pipelines. Both frame-inertial and event-assisted methods retain low errors under this simulated degradation.

RAMPVO is accurate in clear scenes and on Stone blur, but has larger errors in dark and HDR scenes. A mismatch between its training distribution and underwater appearance or event statistics is one possible explanation for this pattern.

These observations motivate evaluating low light and scattering separately, since the two regimes can produce different method rankings.

C. Simulation-to-Real Stability

Figure 4 characterizes ESVIO across three runs per condition. All synthetic repeats produce valid trajectories across clear, blur, dark, and HDR conditions, with errors increasing in dark and HDR scenes.

Valid ESVIO runs are competitive on Cross1, Cross2, and Dark2, but repeated-run status reveals run-to-run instability and divergence on Cross1, Loop2, Dark1, and Blur. RAMPVO and OKVIS2 show related but different synthetic-to-real behavior: RAMPVO is less dominant in simulation but remains competitive on some real sequences, whereas OKVIS2 is accurate in several synthetic clear and blur cases but diverges on multiple AquaticVision sequences. This variability makes it important to consider failed runs alongside errors from successful trajectories when assessing suitability for field use.

Simulation and real data provide complementary evidence. Simulation offers clean ground truth and controlled visibility profiles, while real sequences capture practical calibration, timing, lighting, and event-noise effects. Together, they support analysis of trajectory accuracy and tracking consistency across underwater conditions.

D. Benchmark Use and Extensions

SynUW-VIBench organizes synchronized sensor streams and simulator ground truth around explicit visibility profiles. The brightness statistics and sensor visualizations provide context for interpreting trajectory errors and tracking outcomes. Figure 5 illustrates how the controlled synthetic illumination relates to the mixed lighting observed in AquaticVision, supporting complementary interpretation of the two evaluations.

VII. Conclusion

We presented SynUW-VIBench, a synthetic underwater benchmark pipeline for frame- and event-camera odometry under controlled visibility conditions. With synchronized stereo frames, stereo events, IMU, and ground truth across clear, blur, dark, and HDR scenes, it enables diagnostic comparison of sensing pipelines.

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